

Surface Integrals.

→ Any integral which is to be evaluated over a surface is called a surface integral.

→ Let R be the orthogonal projection of S on xy -plane.

Let $\hat{n} = \cos\alpha \hat{i} + \cos\beta \hat{j} + \cos\gamma \hat{k}$

Then,

$$\iint_S \vec{F} \cdot \hat{n} \, ds = \iint_R \vec{F} \cdot \hat{n} \left(\frac{dx \, dy \, dz}{|\hat{k} \cdot \hat{n}|} \right) =$$

Volume Integrals

Any integral which is to be evaluated over a volume is called a volume integral.

If V is a volume bounded by a surface S , then the triple integrals,

$$\iiint_V \phi \, dv \text{ and } \iiint_V \vec{F} \, dv$$

are called volume integrals.

Q. Evaluate $\iint_S \vec{A} \cdot \hat{n} \, ds$, where $\vec{A} = (x+y^2)\hat{i} - 2xy\hat{j} + 2yz\hat{k}$ and S is the surface of the plane $2x+y+2z=6$ in the first octant?

A vector normal to the surface is,

$$\nabla (2x+y+2z) = 2\hat{i} + \hat{j} + 2\hat{k}$$

$\hat{n}$ = unit vector normal to the surface S

$$= \frac{2\hat{i} + \hat{j} + 2\hat{k}}{\sqrt{4+1+4}}$$

$$= \frac{1}{3} (2\hat{i} + \hat{j} + 2\hat{k})$$

$$\hat{k} \cdot \hat{n} = \hat{k} \cdot \frac{1}{3} (2\hat{i} + \hat{j} + 2\hat{k})$$

$$= \frac{2}{3}$$

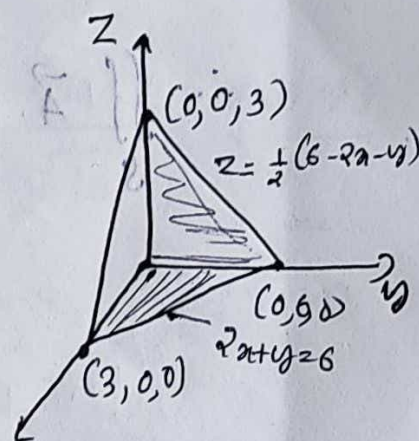
$$\therefore \iint_S \vec{A} \cdot \hat{n} \, ds = \iint_R \vec{A} \cdot \hat{n} \cdot \frac{dxdy}{|\hat{k} \cdot \hat{n}|}$$

where R is the projection of S on the xy -plane. The region R is bounded by x -axis, y -axis and the line $2x+y=6, z=0$.

$$\vec{A} \cdot \hat{n} = (x+y^2)\hat{i} - 2xy\hat{j} + 2yz\hat{k} \cdot \left(\frac{2}{3}\hat{i} + \frac{1}{3}\hat{j} + \frac{2}{3}\hat{k}\right)$$

$$= \frac{2}{3}(x+y^2) - \frac{2}{3}xy + \frac{4yz}{3}$$

$$= \frac{2}{3}x + \frac{2}{3}y^2 - \frac{2}{3}xy + \frac{4yz}{3} = \frac{2}{3}y^2 + \frac{4yz}{3}$$



$$\vec{A} \cdot \hat{n} = \frac{2}{3}y^2 + \frac{4}{3}y \left(\frac{6-2n-y}{2} \right)$$

$$= \frac{2}{3}y \left[y + \frac{6-2n-y}{2} \right]$$

$$= \frac{2}{3}y(6-2n) = \frac{4}{3}y(3-n)$$

$$\iint_S \vec{A} \cdot \hat{n} \, ds = \iint_R \vec{A} \cdot \hat{n} \cdot \frac{dn \, dy}{|\hat{k} \cdot \hat{n}|}$$

$$= \iint_R \frac{4}{3}y(3-n) \cdot \frac{3}{2} \cdot \frac{1}{\sqrt{1+n^2}} \, dn \, dy$$

$$= 2 \iint_R y(3-n) \, dy \, dn$$

$$= 2 \int_0^3 \int_0^{6-2n} y(3-n) \, dy \, dn$$

$$= 2 \int_0^3 (3-n) \left(\frac{y^2}{2} \right)_0^{6-2n} \, dn$$

$$= \int_0^3 (3-n)(6-2n) \, dn$$

$$= 4 \int_0^3 (3-n)^2 \, dn = 4 \left[\frac{(3-n)^3}{-1} \right]_0^3$$

$$= - \left[0 - 3^3 \right] = 81$$

Q. Evaluate $\iint_S \vec{A} \cdot \hat{n} \, ds$, where $\vec{A} = 2xz\hat{i} + xy\hat{j} - 3y^2z\hat{k}$ and S is the surface of cylinder $x^2 + y^2 = 16$ included in the first octant $z=0$ and $z=5$?

$$\nabla(x^2 + y^2) = 2x\hat{i} + 2y\hat{j}$$

$$(\hat{n}) = \frac{2x\hat{i} + 2y\hat{j}}{\sqrt{4x^2 + 4y^2}} = \frac{2x\hat{i} + 2y\hat{j}}{2\sqrt{x^2 + y^2}} = \frac{x\hat{i} + y\hat{j}}{\sqrt{16}} = \frac{x\hat{i} + y\hat{j}}{4}$$

$$|\hat{n}| = 1$$

$$\vec{A} \cdot \hat{n} = \frac{2xz}{4} + \frac{xy}{4} = \frac{1}{4}(2xz + xy)$$

$$\iint_S \vec{A} \cdot \hat{n} \, ds = \iint_R \frac{1}{4}(2xz + xy) \, dy \, dz$$

$$= \iint_R (xz + \frac{1}{4}xy) \, dy \, dz$$

$$= \int_0^5 \int_0^4 (xz + \frac{1}{4}xy) \, dy \, dz = \int_0^5 [xz y + \frac{1}{8}x y^2]_0^4 \, dz$$

$$= \int_0^5 (4xz + \frac{1}{2}x y^2) \, dz$$

$$= [4xz^2 + \frac{1}{2}x y^2 z]_0^5$$

$$= 40 + 2 \times 25$$

$$= 90$$